

NIMA AHMADI, Ph.D.

Rensselaer Polytechnic Institute
Department of Industrial and Systems Engineering

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(Willing to relocate)

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[LinkedIn](#) | [Google Scholar](#) | [Professional Portfolio](#)

Education and Formal Training

- Postdoctoral fellow, Houston Methodist (HM), Houston, TX, USA, 2020 – 2022
- Ph.D., Engineering Management, Western New England University, Springfield, MA, USA, 2015 – 2020
- M.S., Engineering Management, Western New England University, Springfield, MA, USA, 2014 – 2015
- B.Sc., Mechanical Engineering, Azad University Central Tehran Branch, Tehran, 2001 – 2006

Research Interests

- Healthcare human factors, patient safety, and clinical workflow analysis
- Transportation and aviation human factors, training, behavior, and performance assessment
- Physiological and behavioral sensing, including eye tracking, ECG, EEG, and electrodermal activity
- Immersive and simulation environments for assessing stress, workload, behavior, and performance
- Data science, artificial intelligence, and machine learning for human performance modeling and decision support

Academic Appointments and Research Experience

Faculty Member, Rensselaer Polytechnic Institute, 2022-Present

- **Research focus and contributions:** Lead a multidisciplinary collaboration with Albany Medical Center's Emergency Department to analyze vascular access workflows, including procedure duration, catheter utilization, and survival time, using Lean process analysis, task analysis, survival analysis, and ergonomic assessment methods.
- Design and analyze multimodal aviation studies that integrate eye tracking, flight performance data, and voice-based stress indicators; apply machine learning/deep learning methods to evaluate stress and performance among novice general aviation pilots.
- Collaborate with Houston Methodist clinical and research teams to analyze and report findings from a mental health and well-being intervention for healthcare workers, focusing on stress, emotional well-being, workload, and engagement.
- Mentor multidisciplinary capstone design teams and support industry-sponsored engineering design projects, including assistive technology projects aimed at improving mobility, accessibility, and user experience for people with physical disabilities.
- Apply discrete-event simulation, statistical modeling, and data analytics to evaluate ICU and hospital workflows, staffing levels, clinician workload, patient flow, resource utilization, and overall healthcare system performance.
- Teach and develop undergraduate and graduate courses in human factors, discrete-event simulation, design of experiments, supply chain systems, and probability/statistics, incorporating LLM-based tools for report development, data analysis support, and engineering communication.

Postdoctoral Fellow, Houston Methodist, 2020-2022

- **Research focus and contributions:** Contributed to research on tele-critical care, family visitation, clinician mental health, patient engagement, and tailored digital health interventions.
- Participated in the UX research, development, and evaluation of a digital health app for African American breast cancer patients; conducted user interviews, performed thematic analysis, supported formative and validation usability studies, and communicated findings to clinical and project stakeholders.
- Evaluated a multimodal patient communication system using live chat, text messaging, and reminder workflows to assess outpatient appointment adherence, cancellations, rescheduling, and operational performance.
- Designed and coordinated human factors studies in ICU settings, including a prospective longitudinal mixed-methods study of occupational stress and burnout among ICU nurses in COVID and non-COVID units; collected wearable physiological, survey, and qualitative data and contributed to protocol execution, analysis, and reporting.
- Served as a primary point of contact for participant recruitment, protocol execution, data collection, manuscript preparation, stakeholder communication, and research proposal development.
- Developed data pipelines integrating physiological, ocular, behavioral, survey, and operational data for statistical modeling, visualization, and stakeholder reporting.
- Collaborated with cross-functional clinical, engineering, and data science teams and communicated findings through manuscripts, presentations, reports, and stakeholder discussions.

Research Assistant, Western New England University, 2014-2020

- **Research focus and contributions:** Contributed to development and assessment of the Engaged Driver Training System and to development of experimental driving scenarios in the STISIM Drive simulator.
- Participated in longitudinal data collection with more than 40 teen drivers evaluated in simulator-based driving scenarios and a naturalistic on-road study.
- Collected and analyzed eye-tracking and driving-performance data to evaluate hazard-perception skills and assess near and far transfer of training.

Ongoing Research Collaborations and Projects

- Collaborating with Albany Medical Center research teams on midline and central venous catheter placement studies, including task analysis, procedure duration analysis, and workflow assessment.
- Collaborating with Harvard Medical School and RPI Cognitive Science researchers on AI applications for medical emergency cases.
- Developing multimodal research programs using wearable sensors, eye tracking, voice analysis, simulation, and machine learning to evaluate stress, workload, training, and performance.

Teaching Experience

Independent Instructor (2022-Present): Rensselaer Polytechnic Institute, Troy, NY

1. *ISYE 4260: Human Performance Modeling and Support*, undergraduate level
Designed and taught an applied human-factors course integrating sensory, motor, and cognitive performance with multivariate analysis and machine learning. Technical methods included linear and logistic regression, ANOVA/ANCOVA, repeated-measures ANOVA (RM-ANOVA), mixed-design ANOVA, mixed-effects models, and decision-tree classification of human activity. Applied cases addressed eye tracking for concussion detection, gaze and workload assessment, Empatica wearable sensing, virtual reality, and flight simulation. Covered relevant OSHA requirements and human-engineering design guidance in MIL-STD-1472G. Developed Excel exercises and Google Colab notebooks for in-class data analysis and classification activities. **Student rating: 4.8/5.0.**
2. *ISYE 4290/6620: Discrete-Event Simulation*, undergraduate and graduate levels
Taught discrete-event and Monte Carlo simulation for healthcare, manufacturing, service, and risk-analysis applications. Technical topics included process-flow modeling, random-number and random-variate generation, queueing networks, input-distribution fitting, verification and validation (V&V), estimating the required number of replications for terminating and nonterminating systems, determining warm-up periods for nonterminating systems, confidence-interval and output analysis, line balancing, and scenario comparison. Students applied ANOVA to evaluate differences among simulation scenarios and used the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) for multi-criteria ranking of alternatives. They built FlexSim models and used Experimenter/Optimizer, Excel, Python, Minitab, and instructor-developed Google Colab notebooks to improve layouts, throughput, resource utilization, and system performance. **Student rating: 4.7/5.0.**
3. *ISYE 4330/6020: Design of Experiments*, undergraduate and graduate levels
Taught experimental planning, randomization, replication, blocking, one- and two-factor ANOVA, full and fractional factorial designs, confounding, Plackett-Burman screening designs, fixed-, random-, and mixed-effects models, and Response Surface Methodology (RSM). Students used Minitab to estimate main and interaction effects, assess model adequacy, apply stepwise regression, forward selection, and backward elimination for model reduction, optimize processes through the Desirability Function Approach, and evaluate experimental designs reported in peer-reviewed research. Developed Excel demonstrations and Google Colab activities to reinforce ANOVA, factorial effects, model selection, and response-surface visualization. **Student rating: 4.7/5.0.**
4. *ISYE 4210/6600: Design and Analysis of Supply Chains*, undergraduate and graduate levels
Taught quantitative supply-chain design and operations, including time-series forecasting, aggregate planning, economic order quantity, quantity discounts, safety stock, newsvendor models, multi-location inventory, supply-chain coordination, facility-location and set-covering models, transportation, routing, and network design. Students formulated linear and integer optimization models in Excel Solver and Python, analyzed cost-service tradeoffs, and completed applied supply-chain projects; graduate students also critically evaluated peer-reviewed research. **Student rating: 4.2/5.0.**
5. *ISYE 4270/ECSE 4900/MTLE 4920: Multidisciplinary Capstone Design*, undergraduate level
Provided technical and project leadership as Chief Engineer and Faculty Mentor for multidisciplinary student teams completing industry-sponsored engineering design projects. Mentored the Eaton Distribution Center Layout Optimization team that received the RPI Design Lab Performance Excellence Award for Outstanding Achievement (Spring 2026). Guided project planning, requirements development, sponsor engagement, design reviews, analysis, testing, and technical communication while coaching students through technical and teamwork challenges.
6. *ENGR 2600: Modeling and Analysis of Uncertainty*, undergraduate level
Taught descriptive statistics, probability axioms, conditional probability and Bayes' theorem, discrete and continuous random variables, common probability distributions, sampling distributions, the Central Limit Theorem, parameter estimation, confidence intervals, hypothesis testing, paired and two-sample comparisons, chi-square methods, and linear regression. Developed Excel workbooks and Google Colab notebooks for in-class probability, sampling, and statistical-inference activities. **Student rating: 4.4/5.0.**

Independent Instructor (2016-2019): Western New England University, Springfield, MA

1. *IE 212: Probability and Statistics*, undergraduate level; six offerings
Taught engineering probability and statistics, including descriptive measures, set theory, conditional probability, Bayes' theorem, discrete and continuous distributions, mathematical expectation, the Central Limit Theorem, point estimation, confidence intervals, hypothesis tests for means, proportions, and variances, paired comparisons, and linear regression.
2. *IE 419: Industrial Engineering Computer Applications*, undergraduate level; two offerings
Taught applied engineering computing with VBA, Python, SQL, simulation, database methods, and statistical analysis.
3. *ME 205: Measurement Computing*, undergraduate level; three offerings
Taught Visual Basic.NET and VBA for engineering calculations, form and database programming, measurement applications, and data visualization.

Teaching Assistant (Undergraduate Level), Western New England University, Springfield, MA

1. Probability and Statistics Fall 2016
2. Facility Design and Material Handling Fall 2015

Guest Lecturer (Undergraduate Level), Western New England University, Springfield, MA

1. Human Factors Design Engineering Fall 2016
2. Work Analysis and Design Fall 2016
3. Design of Experiments Spring 2016
4. Facility Design and Material Handling Fall 2015

Peer-Reviewed Journal Articles

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- Ahmadi, N., Sasangohar, F., Yang, J., Yu, D., Danesh, V., Klahn, S., & Masud, F. (2024). Quantifying workload and stress in intensive care unit nurses: Preliminary evaluation using continuous eye-tracking. *Human Factors*, 66(3), 714-728. <https://doi.org/10.1177/00187208221085335>
 - Ahmadi, N., Sasangohar, F., Nisar, T., Danesh, V., Larsen, E., Sultana, I., & Bosetti, R. (2022). Quantifying occupational stress in intensive care unit nurses: An applied naturalistic study of correlations among stress, heart rate, electrodermal activity, and skin temperature. *Human Factors*, 64(1), 159-172. <https://doi.org/10.1177/00187208211040889>
 - Ahmadi, N., Romoser, M., & Salmon, C. (2022). Improving the tactical scanning of student pilots: A gaze-based training intervention for transition from visual flight into instrument meteorological conditions. *Applied Ergonomics*, 100, 103642. <https://doi.org/10.1016/j.apergo.2021.103642>
 - Dhala, A., Sasangohar, F., Kash, B., Ahmadi, N., & Masud, F. (2020). Rapid implementation and innovative applications of a virtual intensive care unit during the COVID-19 pandemic: Case study. *Journal of Medical Internet Research*, 22(9), e20143. <https://doi.org/10.2196/20143>
 - Sasangohar, F., Dhala, A., Zheng, F., Ahmadi, N., Kash, B., & Masud, F. (2021). Use of telecritical care for family visitation to ICU during the COVID-19 pandemic: An interview study and sentiment analysis. *BMJ Quality & Safety*, 30(9), 715-721. <https://doi.org/10.1136/bmjqs-2020-011604>
 - Schaefer, E. J., Bracey, A., Ahmadi, N., Brecheisen, J., Rehman, M., Chang, A. K., Ata, A., & Waxman, M. J. (2025). Comparison of procedural duration among ultrasound-guided IVs, mini-midline catheters, and central venous catheters inserted in the emergency department. *Journal of Vascular Access*. Advance online publication. <https://doi.org/10.1177/11297298251395137>

Manuscripts Under Review

- Ahmadi, N., Nisar, T., Shomaker, R., Sasangohar, F., Sheth, P., & Montgomery, T. A multi-modal communication system incorporating live chat to improve outpatient appointment compliance in a large healthcare setting: A quasi-experimental study. Under review, *International Journal for Quality in Health Care*.
- Ahmadi, N., Sasangohar, F. Positive psychological interventions in practice: Effects of a mobile app on emotional experience and mental health in healthcare workers. Under review, *IIE Transactions on Occupational Ergonomics and Human Factors*.
- Ahmadi, N., Romoser, M. Multimodal machine learning assessment of instrument monitoring training and vocal stress in general aviation pilots. Under review, *IEEE Transactions on Human-Machine Systems*.

Manuscripts in Preparation

- Ahmadi, N., Larsen, E., Rao, A., Danesh, V., Faisal, M., & Sasangohar, F. A novel protocol for high-fidelity observation of stress in intensive care nursing through wearable technology.
- Ahmadi, N., Yang, J., Sasangohar, F., Danesh, V., Denney, Y., & Masud, F. Exploring the effect of a Code Blue event on intensive care unit nurses stress and physiological responses: A naturalistic case study.
- Ahmadi, N. Understanding the task structure of vascular access procedures using hierarchical task analysis.
- Ahmadi, N., Katrahmani, A., & Romoser, M. Commentary driving assessment on a tablet-based training program.

Conference Proceedings

- Ahmadi, N., Katrahmani, A., & Romoser, M. (2018). Short- and long-term transfer of training in a tablet-based teen driver hazard perception training program. Proceedings of the Human Factors and Ergonomics Society Annual Meeting, 62, 1965-1969. <https://doi.org/10.1177/1541931218621445>
- Ahmadi, N., Romoser, M., Guarnieri, L., Kry, T., & Porter-Fyke, E. (2018). Application of eye tracking technology in naturalistic usability assessment of an academic library website. *Advances in Usability, User Experience and Assistive Technology*. Springer.
- Katrahmani, A., Ahmadi, N., & Romoser, M. (2017). Using situation awareness as a measure of driver hazard perception ability. 9th International Driving Symposium on Human Factors in Driver Assessment, Training and Vehicle Design. <https://doi.org/10.17077/drivingassessment.1644>
- Demirtas, M., Ahmadi, N., & Dehghanimohammadabadi, M. (2016). Highlighting the main factors of internet banking via multiple criteria decision analysis. Proceedings of the Industrial and Systems Engineering Research Conference. Best Track Paper, Engineering Management Society.
- Zhang, Q., Sasangohar, F., Saravanan, P., Ahmadi, N., Nisar, T., Danesh, V., & Masud, F. (2022). Real-time stress monitoring for intensive care unit (ICU) nurses. Proceedings of the Human Factors and Ergonomics Society Annual Meeting, 66(1), 779-782.
- Zhang, Q., Sasangohar, F., Saravanan, P., Ahmadi, N., Nisar, T., Danesh, V., & Masud, F. (2023). Exploring cognitive stress-related tasks of intensive care unit nurses: A comparison study. Proceedings of the Human Factors and Ergonomics Society Annual Meeting, 67(1), 2205-2209. <https://doi.org/10.1177/21695067231192711>
- Vahdat, V., Salehi, S., & Ahmadi, N. (2019). Service quality assessment via enhanced data-driven MCDM model. In H. Yang & R. Qiu (Eds.), *Advances in Service Science: Proceedings of the 2018 INFORMS International Conference on Service Science* (pp. 217-230). Springer. https://doi.org/10.1007/978-3-030-04726-9_22

Selected Conference Presentations

- Ebnali Harari, M., Ahmadi, N., Al-Taweel, A., & Shokoohi, H. (2023). Clinician-AI collaboration for decision support in telemedicine. International Multi-Disciplinary Conference on Cognitive Methods in Situation Awareness and Decision Support (CogSIMA).
- Ahmadi, N., & Sasangohar, F. (2022). Taming no-show rate: Exploring the impact of facilitating communication with patients using a live chat system and text reminders on patients appointment compliance. Human Factors and Ergonomics Society Annual Meeting.
- Ahmadi, N., Yang, J., Sasangohar, F., Danesh, V., Denney, Y., & Masud, F. (2022). Application of eye movement metrics in naturalistic assessment of intensive care unit nurses mental workload. Human Factors and Ergonomics Society Annual Meeting.
- McNulty, D., Sasangohar, F., Jones-Wood, S., & Ahmadi, N. (2021). Evaluating the effectiveness of virtual mindfulness-based stress reduction. Healthcare Systems Process Improvement Conference.
- Ahmadi, N., Sheth, P., Montgomery, T., & Sasangohar, F. (2021). The impact of a web-based scheduling and communication system on the rate of broken appointments. Healthcare Systems Process Improvement Conference.
- Ahmadi, N., & Sasangohar, F. (2021). Insight into the application of tele-critical care for family visitation in the COVID-19 pandemic. Healthcare Systems Process Improvement Conference.
- Jahroumi, R., Ahmadi, N., Sasangohar, F., & Danesh, V. (2021). Workload intensity in the intensive care unit: Naturalistic physical activities assessment using accelerometer data. Healthcare Systems Process Improvement Conference.
- Ahmadi, N. (2020). Improving the tactical scanning of student pilots. TRB Annual Meeting, ACRP AVO90(1) Subcommittee for Aviation Safety Emerging Research.

Grants and Funded Research

- BEGI Grant, \$10,000. Study of the impact of COVID-19 on ICU nurses through physiological data collection and nurse interviews.

Webinars and Workshops

- Invited Webinar: Tobii Pro webinar on applications of eye-tracking technology across domains, December 2021.
- Workshop Instructor: Summer machine learning workshop for EnMed students, 2020, 2021, and 2023.

Industry Experience

Project Manager and Coordinator, Piping Department, APEC, 2010-2014

- Managed and coordinated basic and detailed piping-engineering activities for water-treatment and steelmaking plant projects.
- Prepared and reviewed piping material specifications (PMS), line lists, piping and equipment layouts, material take-offs, and technical requisitions.
- Used AVEVA PDMS for three-dimensional plant and piping layout, equipment arrangement, routing, and design coordination.
- Applied ASME B31.3 requirements to process-piping design and relevant NFPA standards to firefighting-system design and documentation.

Manager, Heat Exchanger Department, COOLSUMMER, 2009-2010

- Led thermal sizing, rating, selection, and technical evaluation of plate heat exchangers based on required heat duty, flow rates, operating temperatures, pressure drop, and design pressure.
- Selected plate, frame, and gasket materials for different industries based on process-fluid compatibility, temperature, pressure, corrosion resistance, and operating conditions.

- Directed assembly, hydrostatic and leak testing, performance verification, troubleshooting, and repair of plate heat exchangers in accordance with project specifications and applicable requirements.

Journal and Conference Peer Review

- Applied Ergonomics
- IEEE Transactions on Human-Machine Systems
- Winter Simulation Conference

Professional Service and Leadership

- President, American Society for Engineering Management, 2016-2017.
- Member, Graduate Student Advisory Board, Western New England University, August 2017-2020.
- Session Chair, International Conference on Applied Human Factors and Ergonomics (AHFE), July 2018.
- Session Chair, INFORMS International Conference on Service Science, November 2018.
- Chief Engineer and Faculty Mentor, Multidisciplinary Capstone Design, RPI: provided technical leadership, coordinated student teams and industry-sponsored projects, conducted design reviews, and supported sponsor communication and project delivery.
- Faculty Advisor, IISE Student Chapter, Rensselaer Polytechnic Institute; mentored student leaders and supported professional-development activities, industry engagement, and chapter programming.
- Faculty Advisor, Alpha Pi Mu Honor Society, Rensselaer Polytechnic Institute Chapter.
- Member, School of Engineering Generative AI Committee, Rensselaer Polytechnic Institute.
- Department Library Liaison, Industrial and Systems Engineering, Rensselaer Polytechnic Institute.
- Member, ISE Steering Committee; contributed to initiatives in student skill development, staff development, laboratory and equipment planning, and industry-academia partnerships.
- Organized guest lectures across courses by inviting academic and industry professionals to connect technical concepts with current research and engineering practice.
- Advised student simulation projects in manufacturing systems, including warehouse layout optimization, process improvement, and performance analysis.
- Contributor to curriculum development in simulation, design of experiments, human factors, data science, and AI-enabled engineering education.

Professional Affiliations

- Human Factors and Ergonomics Society (HFES), since 2015.
- Institute of Industrial and Systems Engineers (IISE), since 2025.
- Alpha Pi Mu Industrial Engineering Honor Society, since 2025.

Selected Coursework

- **Ph.D. coursework:** Reliability Engineering and Management; Quality Engineering; Statistical Quality Control; Quality Systems and Process Improvement; Design of Experiments; Discrete-Event Simulation; Factory Design and Material Handling; Advanced Optimization Methods; Advanced Production and Inventory Modeling; Engineering Supply Chain; Advanced Risk Analysis; Advanced Cost Estimation; Engineering Psychology; Design of Mechatronic Systems; Energy Management; Seminar and Research Methods for Engineering Management
- **Master's coursework:** Statistical Data Analysis; Advanced Operations Research; Optimization Methods I; Multiple-Criteria Decision Making; Queueing Theory; Stochastic Processes; Design of Experiments; Quality Engineering; Statistical Quality Control; Engineering Cost Analysis; Engineering Risk Management; Engineering Supply Chain

Honors and Awards

- Best Paper Student Competition Award, 3rd Place, Engineering Management Society, 2016, for "Highlighting the Main Factors of Internet Banking via Multiple Criteria Decision Analysis."
- Engineering Research Assistantship Scholarship, Western New England University, 2016.

Software Skills

- **Statistical Analysis:** Minitab, Python, SPSS, MS Excel
- **Simulation and Risk Analysis:** FlexSim, @RISK
- **Programming and Machine Learning:** Python, VBA
- **Optimization:** Excel Solver, Python
- **Data Visualization:** Power BI, Python
- **Generative AI:** OpenAI API and LLM development tools
- **Database and Engineering Software:** MySQL, MongoDB, MS Project, AutoCAD

Certifications and Professional Development

- Six Sigma Green Belt Program, Western New England University, 2017
- Springboard Intermediate Data Science with Python, 2018-2019
- Machine Learning with Python, 2020
- Neural Networks and Deep Learning, 2023